CS & IT ENGINEERING

Operating System

File System

Lecture -2

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Recap of Previous Lecture

doubt ?

Topic File System

Topic Blocks

Topics to be Covered

Topic Blocks and File Allocation Methods

Topic Unix i-node

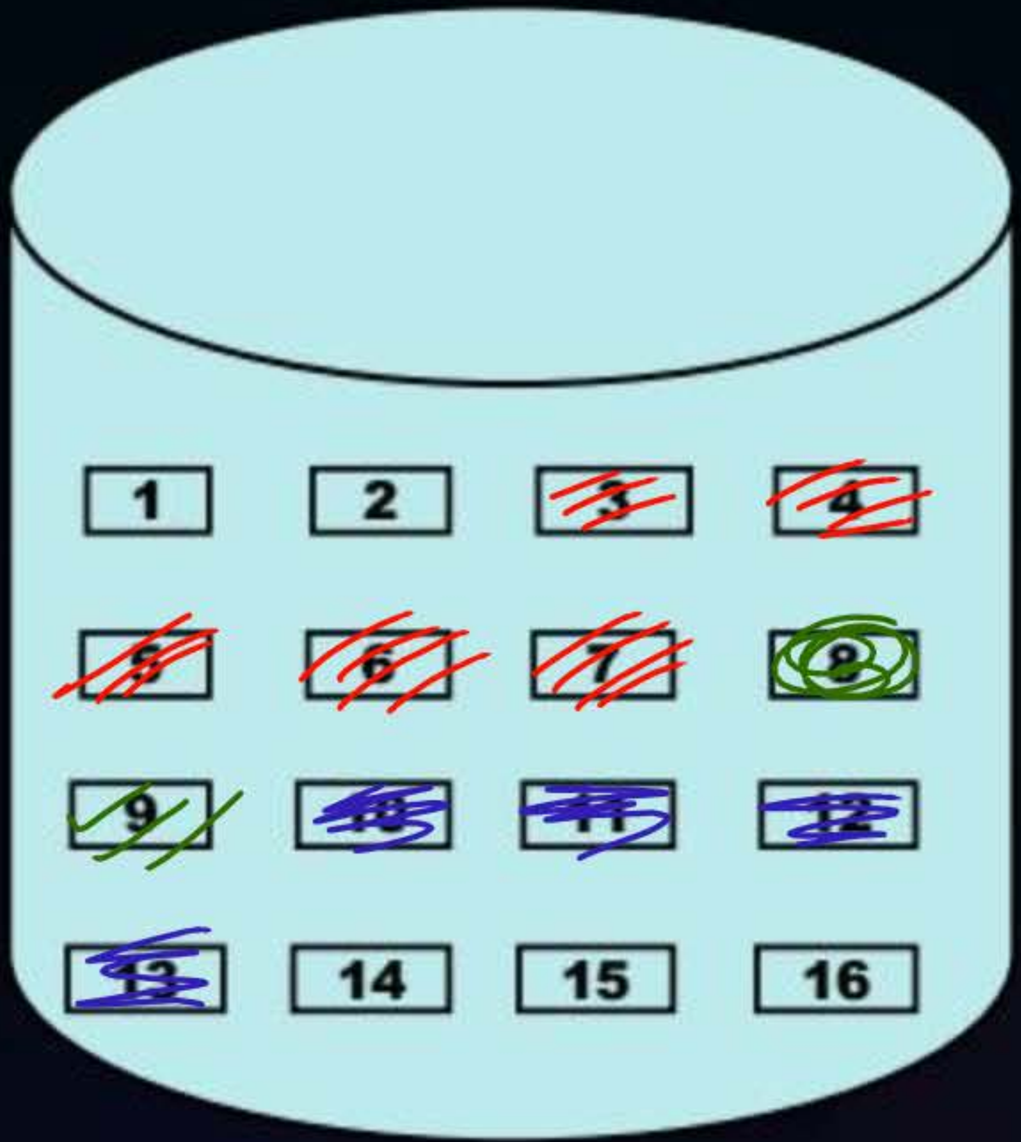
Topic : File Allocation Method

→ how a file allocated on disk blocks

1. Contiguous Allocation
2. Linked Allocation
3. Indexed Allocation

Topic : Contiguous Allocation

file is stored on consecutive blocks



file allocation table

Name	starting block no.	No. of blocks
OS.pdf	3	5
COA.pdf	10	4
CN.pptx	8	1

Topic : Contiguous Allocation

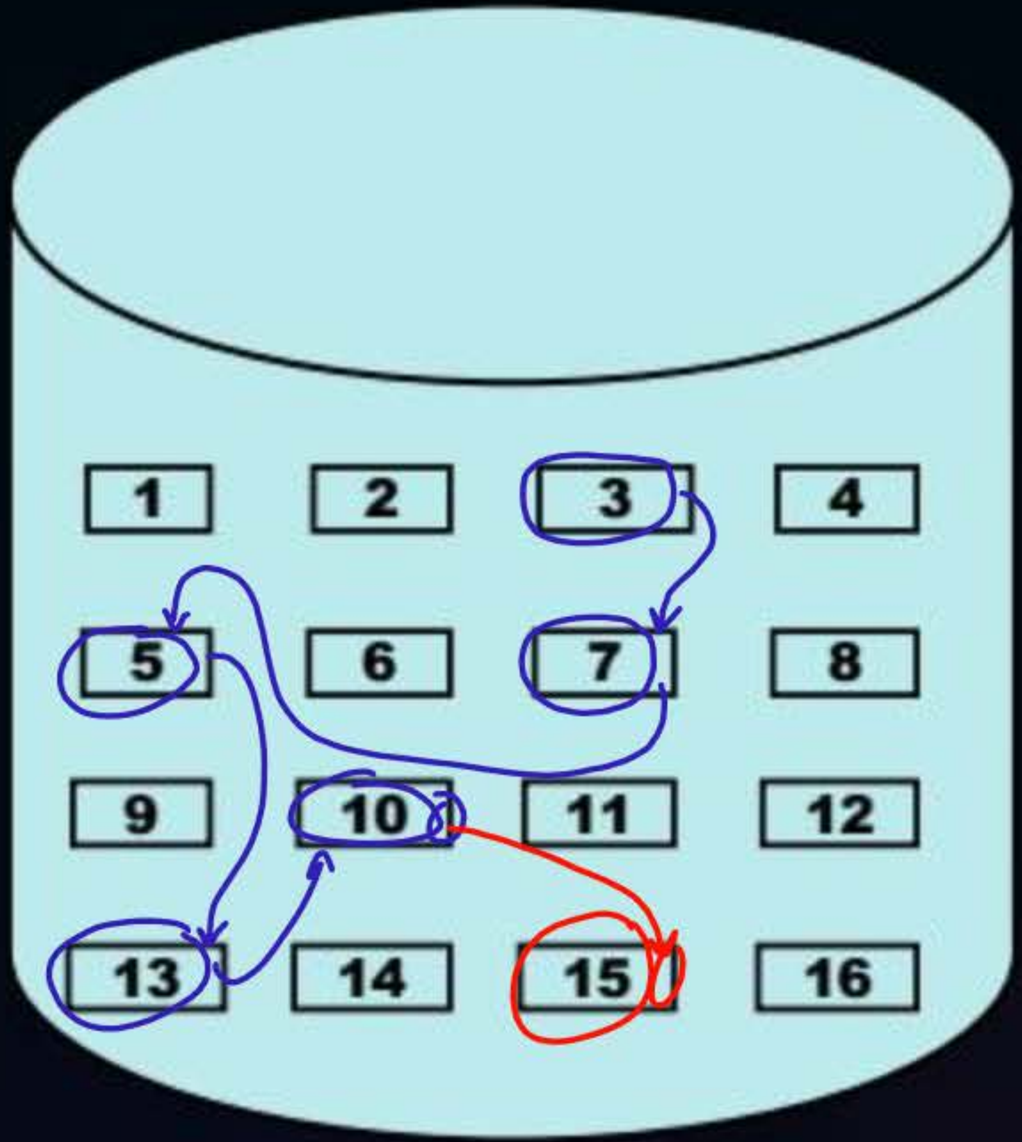
Performance:

1. Fragmentation: Internal, External
2. Increase in File size: Inflexible ← when free consecutive block are not available
3. Type of access: Sequential, Random/direct

Insertion of a new block in file at	No. of block accesses
Beginning	Many
Middle	many
End	Lesser

Topic : Linked Allocation

→ file is allocated on available free blocks like a linked list



file allocation table

Name	starting block no	Last block no.
os.pdf	3	10

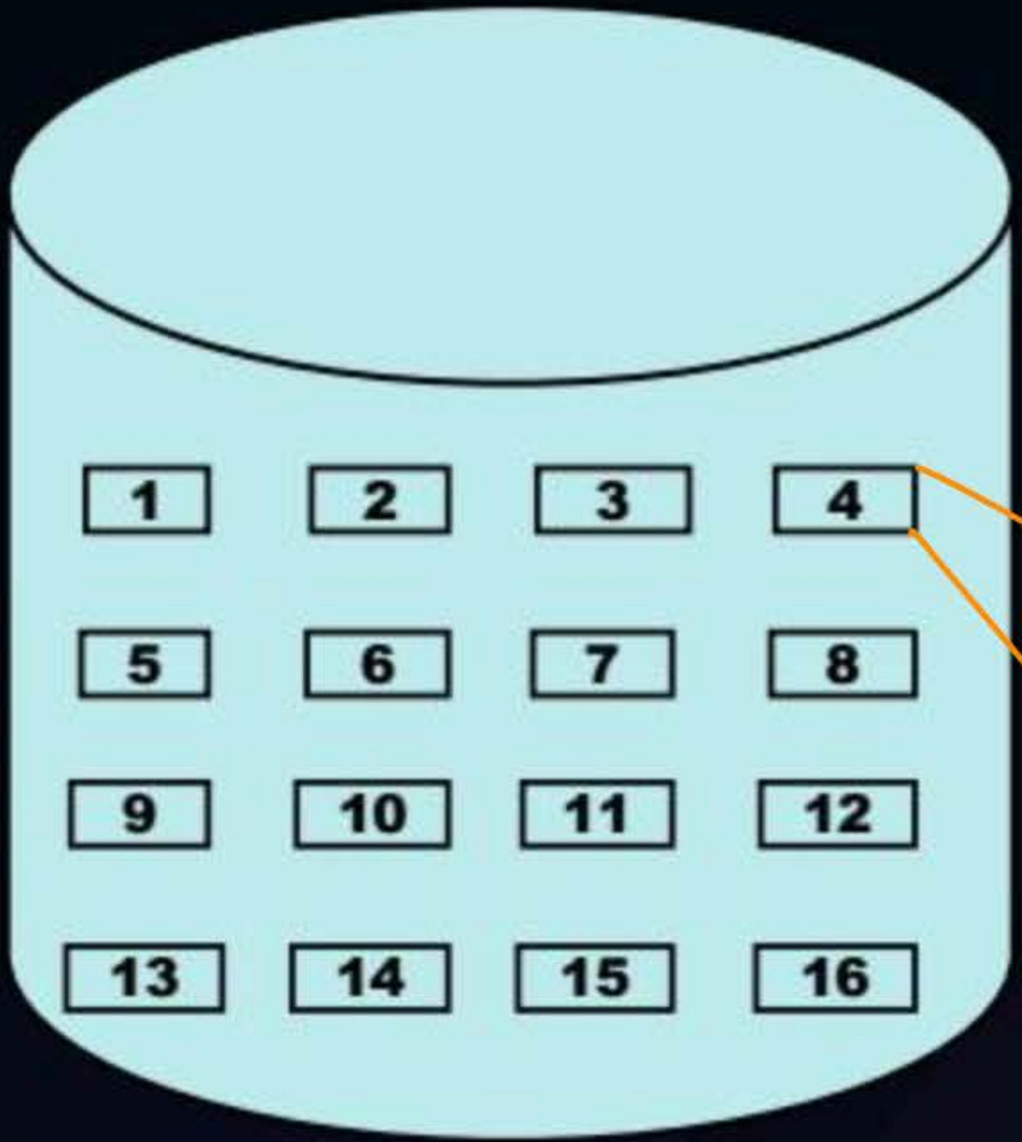
Topic : Linked Allocation

Performance:

1. Fragmentation: Internal
2. Increase in File size: Flexible
3. Type of access: Sequential

Insertion of a new block in file at	No. of block accesses
Beginning	Less
Middle	many
End	Many

Topic : Indexed Allocation



file allocation table

file name	Index block no.
os. pdf	4

Topic : Indexed Allocation

Performance:

1. Fragmentation: Internal
2. Increase in File size: Flexible
3. Type of access: Sequential, Random/direct

Insertion of a new block in file at	No. of block accesses
Beginning	2
Middle	2
End	2

one for new file block
one for index block

Disadv :-

some blocks are occupied for indexes hence, no. of blocks remaining to store file will be reduces

Topic : Question

- Number of Disk blocks = 1024
- Disk block size = 1KB
- [illegible]
- Number of blocks are blocked for indexing = 4
- Maximum file size?

No. of blocks available to store file $= 1024 - 4 = 1020$

max file size $= 1020 * 1KB$

$= \underline{1020\ KB}$ Ans.

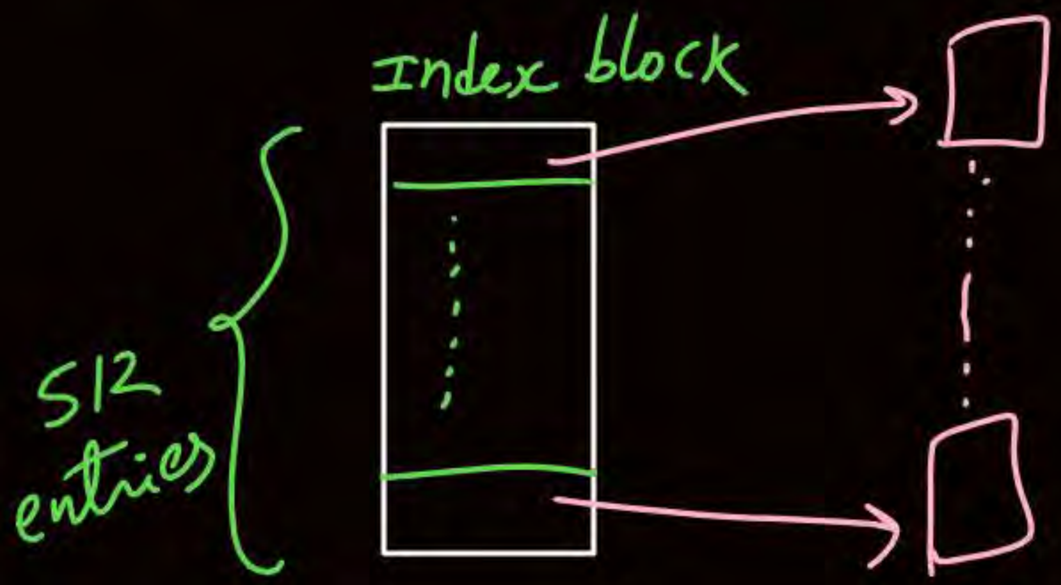
Ques) Disk block size = 2KB

Disk block address = 4B (32 bits)

max no. of indexes stored in one block = 512 ?

max file size possible = 1MB ?

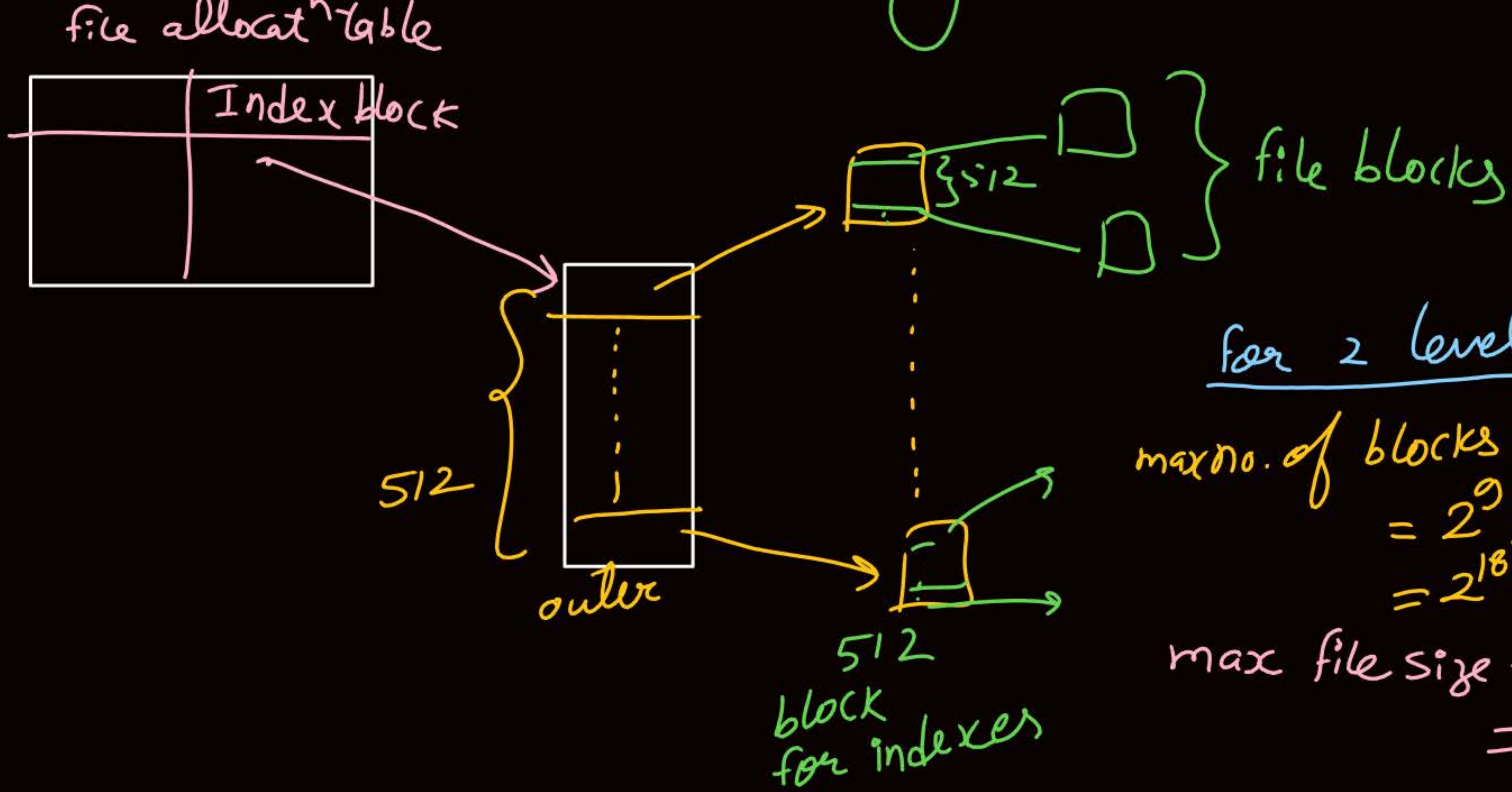
Sol^n no. of indexes per block = $\frac{2KB}{4B} = 2^9 = 512$



max file size = 512 * 2KB
= 1MB

} if indexes are stored only in 1 block.

If more than one blocks are needed to store Indexes
⇓
multilevel indexing



For 2 level indexing:-

max no. of blocks to store file
$= 2^9 * 2^9$
$= 2^{18}$

max file size $= 2^{18} * 2kB$
$= 512 MB$

Topic : Question

H.W.

[GATE-2022]

#Q. Consider two files systems A and B , that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50^{th} and 51^{st} block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B respectively, then the value of $n_A + n_B$ is_______?

Topic : Question

[GATE-2025]

#Q. A disk of size 512M bytes is divided into blocks of 64Kbytes. A file is stored in the disk using linked allocation. In linked allocation, each data block reserves 4 bytes to store the pointer to the next data block. The link part of the last data block contains a NULL pointer (also of 4 bytes). Suppose a file of 1M bytes needs to be stored in the disk. Assume, $1K = 2^{10}$ and $1M = 2^{20}$. The amount of space in bytes that will be wasted due to internal fragmentation is 65468. (Answer in integer)

No. of blocks needed to store file $= \frac{1\,mB}{64\,KB} = \frac{2^{20}}{2^{16}} = 2^{4} = 16$

$\Downarrow$

without pointer

with pointer = 17 blocks

$\Downarrow$

17 pointers

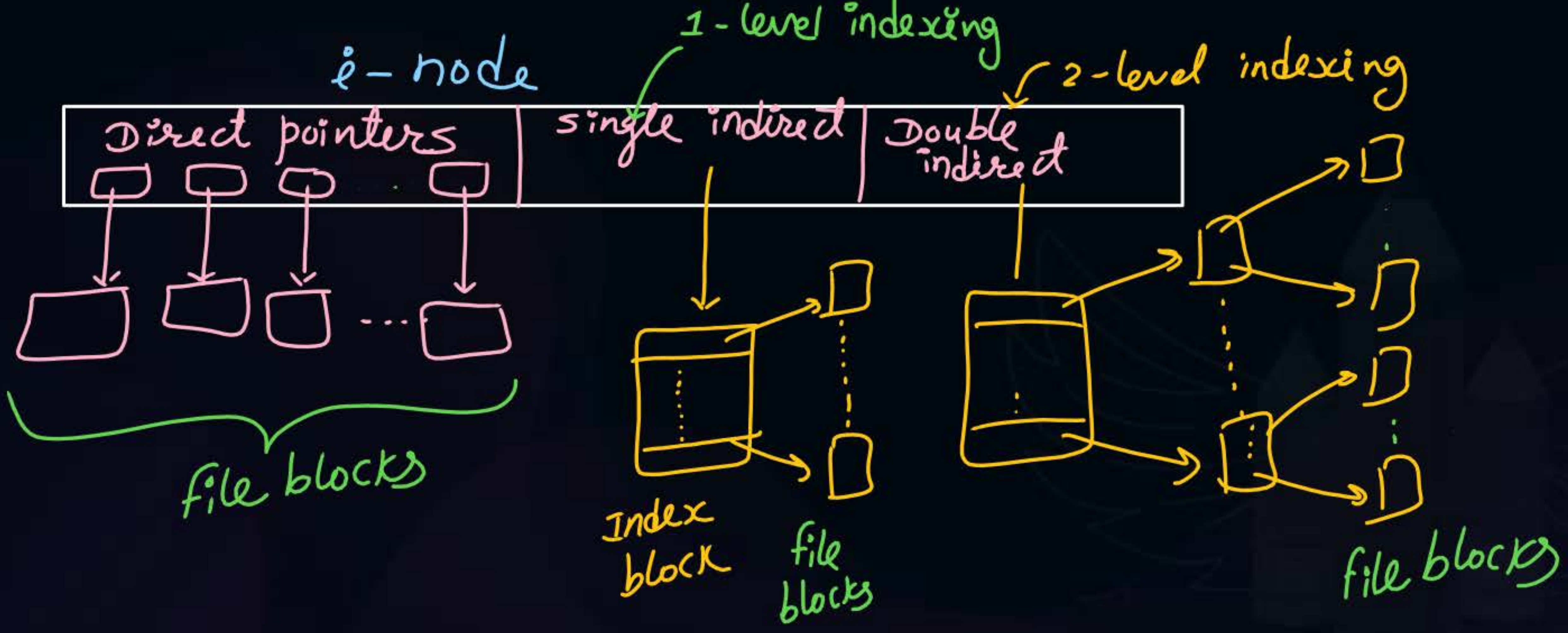
Content on last block $= (16 * 4) + 4 = 68$ bytes

Amount of internal fragmentation $= 65536 - 68$

$= \underline{\underline{65468}}$ Ans.

Topic : Unix I-node Structure

The i-node (index node) is a data structure in a Unix-style file system that describes a file-system object such as a file or a directory.



Topic : Unix I-node Structure

The i-node (index node) is a data structure in a Unix-style file system that describes a file-system object such as a file or a directory.

- Each i-node stores the attributes and disk block locations of the object's data.
- The number of I-node limits the total number of files/directories that can be stored in the file system.

Topic : Question

[GATE-2019]

#Q. The index node (inode) of a Unix-like file system has 12 direct, one single-indirect and one double-indirect pointer. The disk block size is 4 kB, and the disk block addresses 32-bits long. The maximum possible file size is (rounded off to 1 decimal place) ___4.0___ GB?

4B

no. of indexes per block $= \frac{4KB}{4B} = 1K$

max file size $= (12 * 4KB) + (1k * 4KB) + (1k * 1k * 4KB)$

$= 48KB + 4MB + 4GB$

$\simeq 4.004048\,GB$

$\simeq 4.0\,GB$

Topic : Question

H.W.

[GATE-2005]

#Q. In a computer system, four files of size 11050 bytes, 4990 bytes, 5170 bytes and 12640 bytes need to be stored. For storing these files on disk, we can use either 100 byte disk blocks or 200 byte disk blocks (but can't mix block sizes). For each block used to store a file, 4 bytes of bookkeeping information also needs to be stored on the disk. Thus, the total space used to store a file is the sum of the space taken to store the file and the space taken to store the book keeping information for the blocks allocated for storing the file. A disk block can store either bookkeeping information for a file or data from a file, but not both.

What is the total space required for storing the files using 100 byte disk blocks and 200 byte disk blocks respectively?

A. 35400 and 35800 bytes

B. 35800 and 35400 bytes

C. 35600 and 35400 bytes

D. 35400 and 35600 bytes

Topic : Solution

File Size	No. of blocks to store file	No. of blocks to store bookkeeping	Total blocks	Total Size
11050 Bytes				
4990 Bytes				
5170 Bytes				
12640 Bytes				

Topic : Solution

File Size	No. of blocks to store file	No. of blocks to store bookkeeping	Total blocks	Total Size
11050 Bytes				
4990 Bytes				
5170 Bytes				
12640 Bytes				

2 mins Summary

Topic Blocks and File Allocation Methods

Topic Unix i-node

Happy Learning

THANK - YOU